UNDERGRADUATE THERMAL ENGINEERING LABORATORY

SPHERE: MISSION CONTROL

STUDENT MANUAL, EXPERIMENTAL METHOD AND ANALYSIS GUIDE

Experimental objective: Construct an insulated $100\ \mathrm{mL}$ water capsule, expose it to an ice-water boundary ($0^\circ\mathrm{C}$) and record the temperature for $15\ \mathrm{minutes}$. Compare the measurements with Newton's law of cooling using residuals and MAE ($\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\Delta T_i|$).

1. ENGINEERING CONTEXT & OBJECTIVES

Spacecraft thermal control must account for radiative exchange, internal heat generation and highly variable external heat fluxes. In vacuum there is no external convection; thermal design therefore differs fundamentally from this water-bath experiment.

Aerospace multilayer insulation uses low-emissivity reflective layers separated by low-conductance spacers. The cotton, bubble wrap and reflective film used here form an educational multilayer assembly, not flight-qualified MLI.

Test the assigned insulation configuration under a common boundary condition. Compare observations with the analytical reference curve and report residuals, MAE, uncertainty sources and model limitations.

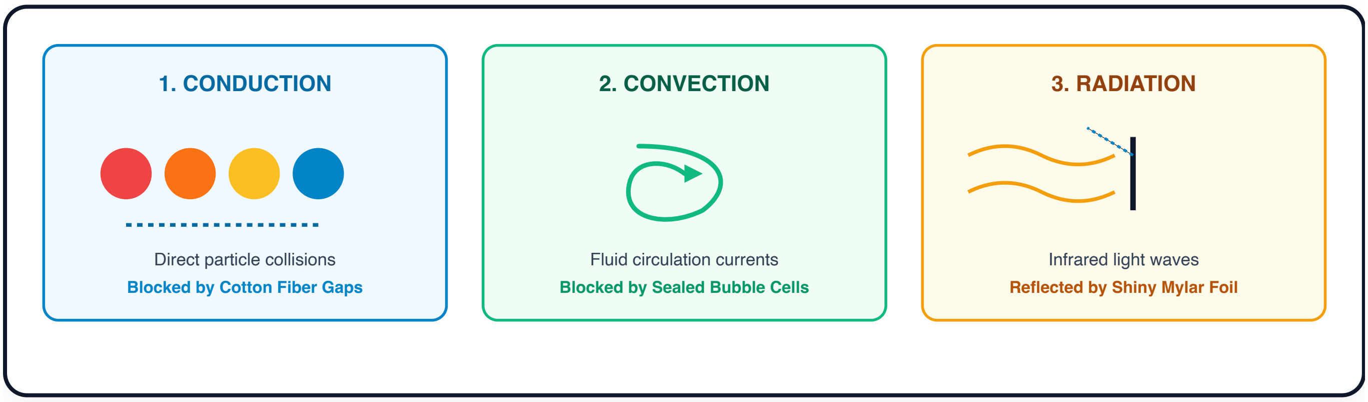


FIGURE 2: THREE MODES OF HEAT TRANSFER & THERMAL INSULATION BARRIERS

2. HARDWARE SETUP & SCHEMATIC

Assemble a sealed test capsule suitable for controlled immersion:

HARDWARE PART	DESCRIPTION	WHAT IT DOES IN OUR LAB
Core Jar	$100\ \mathrm{mL}$ wide-mouth PET jar	Holds our hot water core ($T_0 = 80.0°\mathrm{C}$), representing the astronaut's internal body heat.
Prepared Lid	Jar lid with a probe opening	Supports consistent probe placement; seal using the approved lab method and leak-test before use.
Digital Sensor	Waterproof digital thermometer	Measures internal core water temperature (T_{obs}, accuracy $\pm 0.5°\mathrm{C}$). Record the accuracy for your instrument.
Ice Bath Chamber	Ice-water slurry ($0.0°\mathrm{C}$)	Provides a repeatable low-temperature boundary sink ($T_{\mathrm{env}} = 0.0°\mathrm{C}$) for comparative testing.

Quick Assembly Steps:

1. **Feed the Sensor:** Slide the metal temperature probe through the prepared opening in the jar lid.
2. **Set the Depth:** Position and secure the lead so the probe remains at the required depth.
3. **Seal and Test:** Seal the probe opening using the approved lab method, close the lid, and complete a leak test before immersion.
4. **Center the Probe:** Make sure the metal sensor tip hangs right in the center of the liquid without touching the bottom or sides.

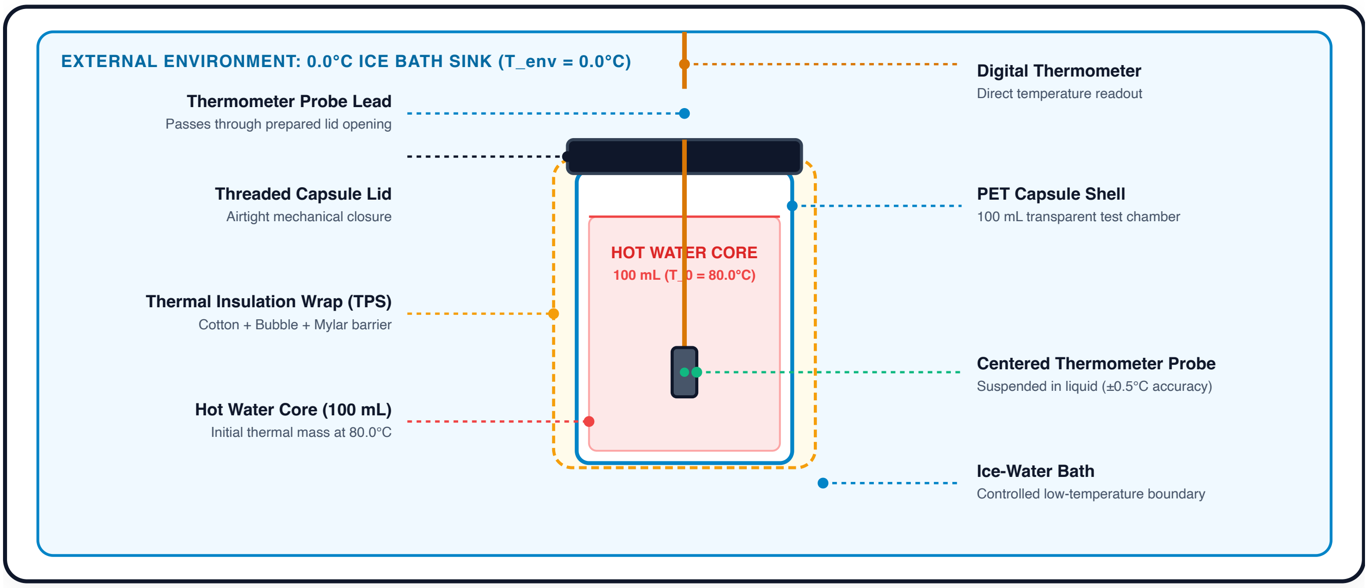


FIGURE 1: EXPERIMENTAL HARDWARE ASSEMBLY & SENSOR ALIGNMENT

3. THE 4 INSULATION CONFIGURATIONS

Each team tests one of four configurations under the same nominal boundary conditions ($T_0 = 80.0°\mathrm{C}, T_{\mathrm{env}} = 0.0°\mathrm{C}$):

A. BARE CONTROL

$k \approx 0.150\ \mathrm{min}^{-1}$

Plain uninsulated jar. Shows how rapidly heat escapes through direct conduction and water currents.

B. BUBBLE WRAP

$k \approx 0.040\ \mathrm{min}^{-1}$

Two even layers of bubble wrap. Sealed air cells reduce fluid motion and slow heat transfer.

C. MYLAR FOIL

$k \approx 0.121\ \mathrm{min}^{-1}$

Shiny emergency blanket. Reflects radiant thermal infrared waves back toward the hot water.

D. MULTILAYER ASSEMBLY

$k \approx 0.015\ \mathrm{min}^{-1}$

Combines cotton, bubble wrap and reflective film to increase effective thermal resistance.

Optional extension: Add a fifth cotton-only trial to isolate the contribution of the inner fibrous layer.

FIGURE 3: EDUCATIONAL MULTILAYER TEST ASSEMBLY

4. THE PHYSICS: NEWTON'S LAW OF COOLING

Heat always flows from warmer objects to cooler surroundings. Over 300 years ago, Sir Isaac Newton discovered that the rate at which an object cools is proportional to the temperature difference between the object and its environment:

NEWTON'S LAW OF COOLING (EXPONENTIAL DECAY MODEL)

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-k \cdot t}$$

Where:

$T(t)$ = Predicted core temperature ($^\circ\text{C}$) at minute t

T_{env} = Ice bath temperature (0.0°C)

T_0 = Starting core temperature (80.0°C)

k = Cooling rate constant (min^{-1})

Interpretation of k: The fitted cooling constant k has units of min^{-1}. Under otherwise identical conditions, a smaller value corresponds to a slower approach to the bath temperature.

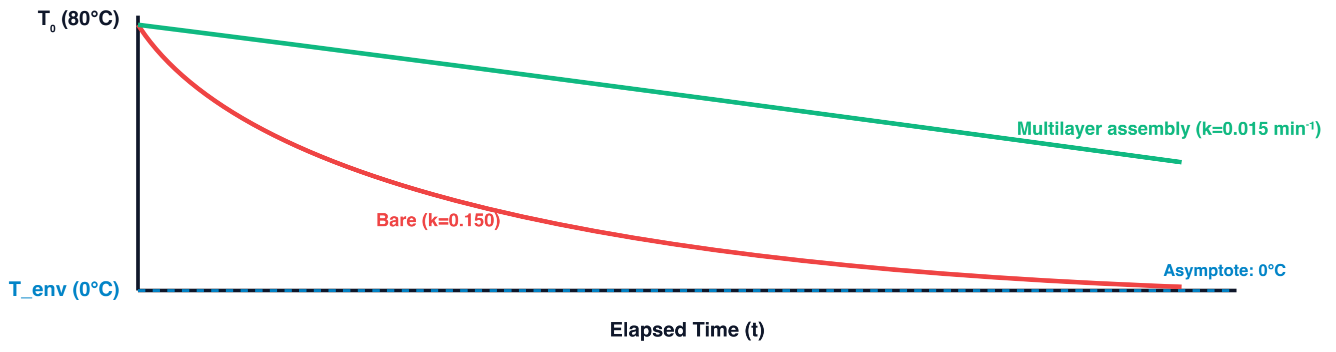


FIGURE 4: EXPONENTIAL DECAY PROFILES ACROSS HIGH VS LOW COOLING CONSTANTS (k)

5. STEP-BY-STEP LAB WORKFLOW

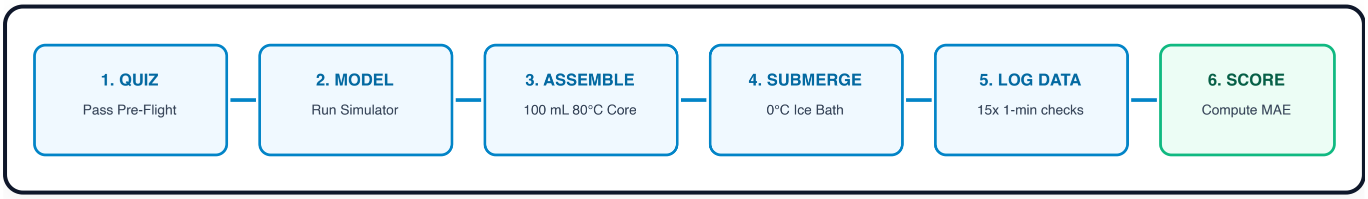


FIGURE 5: 6-STAGE EXPERIMENTAL PROTOCOL WORKFLOW

1. **Complete the concept check:** Review the heat-transfer questions before handling hot water.
2. **Generate Model Prediction:** Go to the **Simulator** tab, choose your team's assigned insulation configuration, verify starting conditions ($T_0 = 80.0°\mathrm{C}$, $T_{\mathrm{env}} = 0.0°\mathrm{C}$), and click **RUN SIMULATION PREDICTION**.
3. **Fill the Water Core:** Carefully measure $100\ \mathrm{mL}$ of hot water at $80.0°\mathrm{C}$, pour it into your jar, screw the cap on tightly, and verify the seal is leak-free.
4. **Submerge and start timing:** Lower the capsule into the measured ice-water bath $(0.0°\mathrm{C})$ and immediately start the timer in the **Measurement Log** tab.
5. **Log Temperature Every Minute:** Every $60\ \mathrm{seconds}$ for $15\ \mathrm{minutes}$, write down the observed temperature in your Task Sheet and enter it into Mission Control.
6. **Evaluate:** Plot the measured and predicted curves, calculate residuals ($|\Delta T| = |T_{\mathrm{obs}} - T_{\mathrm{model}}|$) and MAE $(\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\Delta T_i|)$, and discuss the principal sources of uncertainty.

SECTION 6: STUDENT TASK SHEET & GRAPH WORKBOOK

WORKSHEET #01

STUDENT NAME(S):

TEAM / TEST IDENTIFIER:

DATE / CLASS PERIOD:

TESTED INSULATION CONFIGURATION:

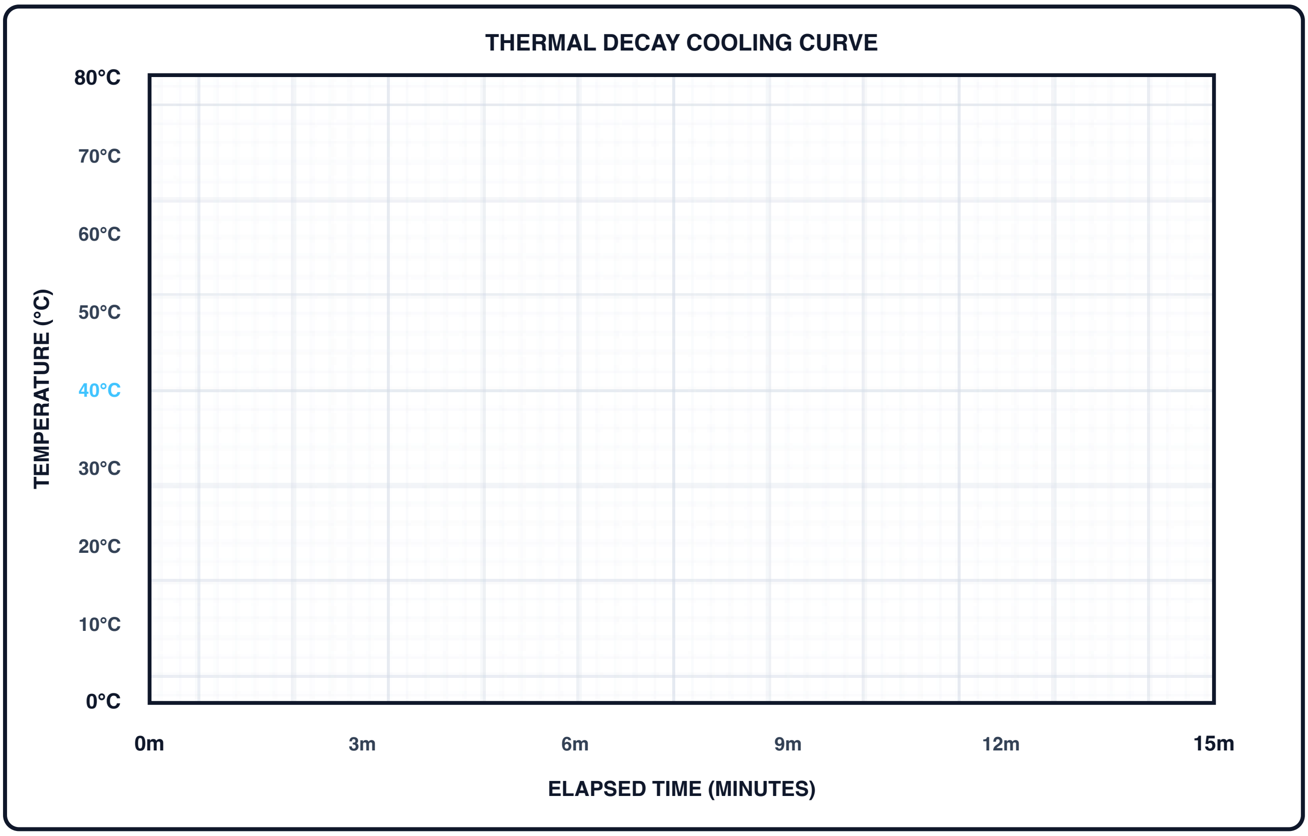
[ ] A: Bare [ ] B: Bubble wrap [ ] C: Reflective film [ ] D: Multilayer

Part A. Experimental Temperature Log

TIME (min)	MEASURED TEMP (T_{obs}, $^\circ\mathrm{C}$)	PREDICTED TEMP (T_{model}, $^\circ\mathrm{C}$)	ABS DIFFERENCE $\lvert\Delta T\rvert$ ($^\circ\mathrm{C}$)	OBSERVER NOTES
0 min	$80.0^\circ\mathrm{C}$	$80.0^\circ\mathrm{C}$	$0.0^\circ\mathrm{C}$	Capsule submerged in $0.0^\circ\mathrm{C}$ ice bath
1 min				Initial cooling rate check
2 min				
3 min				
4 min				
5 min				Mid-point check; check ice bath
6 min				
7 min				
8 min				
9 min				
10 min				Check curve flattening
11 min				
12 min				
13 min				
14 min				
15 min				Final measurement checkpoint

Part B. Coordinate Graph Grid

Plot your measured physical readings as solid dots **(•)** and connect them with a smooth line. Draw the theoretical model line as a dashed curve **(---)**.



Part C. Residual and Mean Absolute Error Analysis

Calculation Steps:

1. **Sum of absolute residuals:** Add the n valid entries in the $|\Delta T|$ column: $\sum |\Delta T| = \sum_{i=1}^{n} |T_{\text{obs},i} - T_{\text{model},i}| =$ ____________ $^\circ\text{C}$
2. **Mean absolute error:** Divide by the number of valid observations (n): $\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |T_{\text{obs},i} - T_{\text{model},i}| =$ ____________ $^\circ\text{C}$
3. **Interpretation:** A lower MAE indicates closer agreement. Report the value with units ($^\circ\text{C}$) and discuss whether it is meaningful relative to thermometer accuracy ($\pm 0.5^\circ\text{C}$) and repeatability.

REPORT: $\text{MAE} =$ _________ $^\circ\text{C}$ | **NUMBER OF OBSERVATIONS** $n =$ _________

7. LAB ANALYSIS & REFLECTION QUESTIONS

This is the eight-question extended reflection set. The separate two-page task sheet contains six condensed practical inquiries, while Mission Control provides a separate 10-item online knowledge check.

Question 1: Why can reflective film provide limited insulation in direct water contact?

Reflective film can reduce radiative exchange when it faces a suitable gap. Explain why it may provide limited insulation when placed in direct contact with the capsule and ice-water bath without a low-conductivity spacer.

Question 2: What does the cooling constant (k) tell us?

In Newton's Law $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-kt}$, does a *smaller* k value mean a better insulator or a worse insulator? Explain using the formula.

Question 3: How does Multi-Layer Insulation (MLI) work together?

Explain how cotton, bubble wrap and reflective film can provide complementary thermal resistances in Configuration D. Treat $k \approx 0.015\ \text{min}^{-1}$ as an assumed reference value to test, not a guaranteed result.

Question 4: Where could experimental errors come from?

List two physical reasons why your real sensor readings (T_{obs}) might differ from the smooth curve (T_{model}) predicted by the computer model.

Question 5: What happens if your ice bath warms up?

If your classroom ice bath warms from $0.0°\mathrm{C}$ to $4.5°\mathrm{C}$ while testing, what should you change in the Mission Control simulator so the prediction (T_{env}) stays accurate?

Question 6: What if we double the water volume?

If we doubled the water inside the jar from $100\ \mathrm{mL}$ to $200\ \mathrm{mL}$, would it cool down faster or slower? How does thermal energy ($Q = m \cdot c \cdot \Delta T$) relate to this, and would k increase or decrease?

Question 7: Spacesuit Design Trade-Offs

If adding insulation layers keeps astronauts warmer, why don't space engineers just wrap a spacesuit in $50\ \mathrm{layers}$ of MLI? What problems would that cause for an astronaut on a spacewalk?

Question 8: How can measurement agreement be improved?

Describe two practical changes that could reduce MAE in a repeated experiment.

